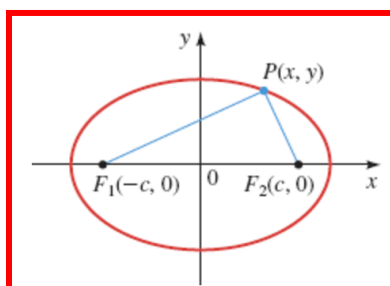
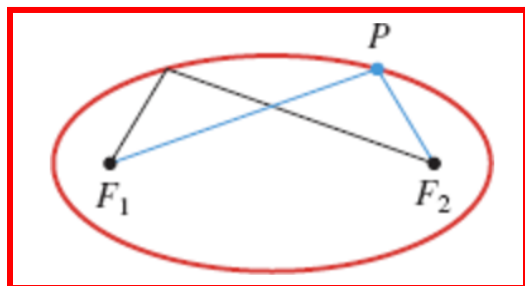


An **ellipse** is an oval curve that looks like an **elongated circle**.

Geometric Definition of an Ellipse

An ellipse is the set of all points in the plane the **sum of whose distances from two fixed points**

F_1 and F_2 is a **constant**. These two fixed points are the **foci** (plural of focus) of the ellipse.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$x^2 = a^2$$

$$x = \pm a$$

$$\frac{y^2}{b^2} = 1$$

$$y^2 = b^2$$

$$y = \pm b$$

$$x^2 + 2y^2 + 6x - 5y + 2 = 0$$

The standard equation of an ellipse with foci on the x -axis and center at the origin is

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$). If we let $y = 0$ then $x = \pm a$. The ellipse crosses the x -axis at $(a, 0)$ and

$(-a, 0)$, as shown below. These points are called the **vertices** of the ellipse, and the segment that

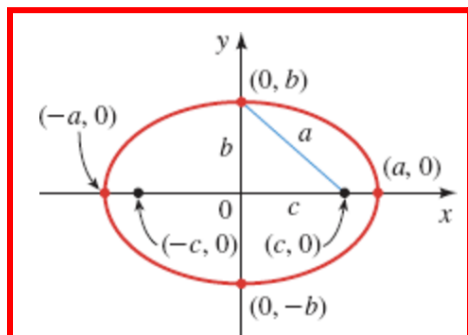
joins them is called the **major axis**. Its length is $2a$. If we let $x = 0$ then $y = \pm b$. The ellipse

crosses the y -axis at $(0, b)$ and $(0, -b)$. The segment that joins these points is called the **minor**

axis, and it has length $2b$. If the foci of the ellipse are placed on the y -axis at $(0, \pm c)$ rather

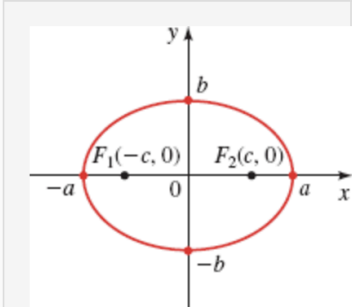
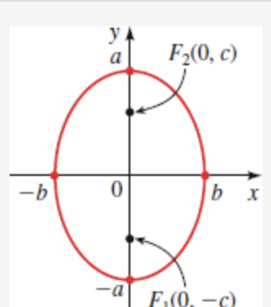
than on the x -axis, then the roles of x and y are reversed in the preceding discussion, and we get a

vertical ellipse.



Ellipse with Center at the Origin

The graph of each of the following equations is an ellipse with center at the origin and having the given properties.

EQUATION	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow x\text{-major}$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \rightarrow y\text{-major}$
	$a > b > 0$	$a > b > 0$
VERTICES	$(\pm a, 0)$	$(0, \pm a)$
MAJOR AXIS	Horizontal, length $2a$	Vertical, length $2a$
MINOR AXIS	Vertical, length $2b$	Horizontal, length $2b$
FOCI	$(\pm c, 0), c^2 = a^2 - b^2$	$(0, \pm c), c^2 = a^2 - b^2$
GRAPH		

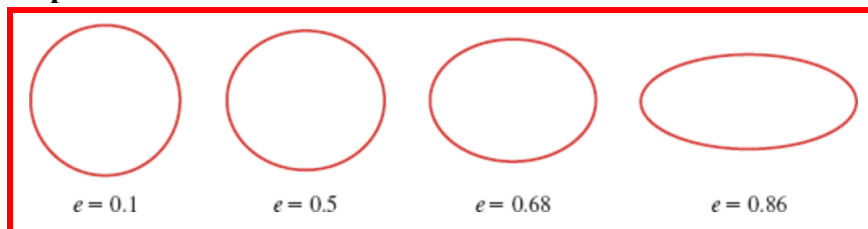
In the standard equation for an ellipse, a^2 is the *larger* denominator, and b^2 is the *smaller*. To find c^2 , we subtract: **larger denominator minus smaller denominator**.

Definition of Eccentricity

For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ or $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ (with $a > b > 0$), the eccentricity e is the number $e = \frac{c}{a}$ where $c = \sqrt{a^2 - b^2}$. The eccentricity of every ellipse satisfies $0 < e < 1$. If $a = b$ in the equation of an ellipse, then $\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$ so $x^2 + y^2 = a^2$. In this case the “ellipse” is a circle with radius a and eccentricity 0 .

Thus if e is close to 1, then c is almost equal to a , and the ellipse is elongated in shape, but if e is close to 0, then the ellipse is close to a circle in shape. The eccentricity is a measure of how “stretched” the ellipse is.

Ellipses with various eccentricities



Example 1: An equation of an ellipse is given.

- Find the vertices, foci, and eccentricity of the ellipse.
- Determine the lengths of the major and minor axes.
- Sketch a graph of the ellipse.

i. $\frac{x^2}{9} + \frac{y^2}{64} = 1$

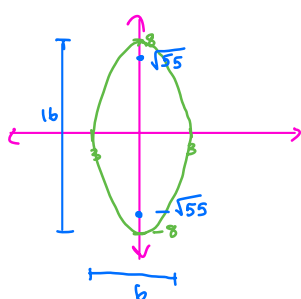
$a = 8$, y-major

$b = 3$

$c = \sqrt{64-9} = \pm\sqrt{55}$

$(0, \pm 8)$ $(\pm 3, 0)$

$e = \frac{c}{a} = \frac{\sqrt{55}}{8}$, $2a = 16$
 $2b = 6$



ii. $3x^2 + 4y^2 = 12 \rightarrow$ x-major

$\frac{1}{4}x^2 + \frac{1}{3}y^2 = 1$

$\frac{x^2}{4} + \frac{y^2}{3} = 1$

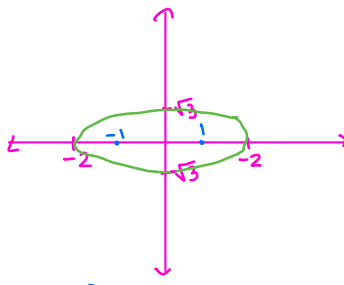
$a = 2$

$b = \sqrt{3}$

$c = \sqrt{2^2-3} = \pm 1$

$(\pm 2, 0)$

$e = \frac{c}{a} = \frac{1}{2}$



iii. $x^2 = 4 - 2y^2$

$\frac{x^2}{4} + \frac{2y^2}{4} = \frac{4}{4}$

$\frac{x^2}{4} + \frac{y^2}{2} = 1 \rightarrow$ x-major

$a = 2$

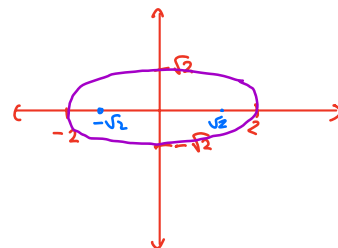
$b = \sqrt{2}$

$c = \sqrt{4-2} = \pm\sqrt{2}$

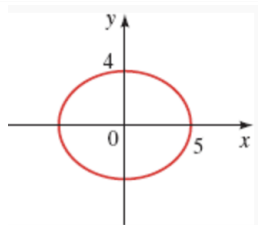
foci: $(\pm\sqrt{2}, 0)$

$2a = 4$, $2b = 2\sqrt{2}$

$e = \frac{c}{a} = \frac{\sqrt{2}}{2}$



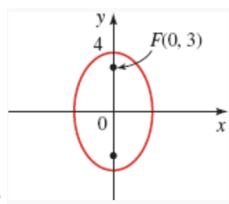
Example 2: Find the standard equation for the ellipse whose graph is shown.



a.

x-major, $a = 5$
 $b = 4$

$\frac{x^2}{25} + \frac{y^2}{16} = 1$



b.

y-major

$c = 3$

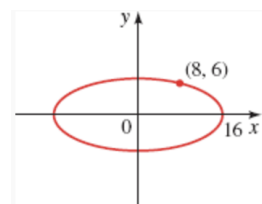
$a = 4$

$3^2 = 4^2 - b^2$

$b^2 = 16 - 9$

$b^2 = 7$

$\frac{x^2}{7} + \frac{y^2}{16} = 1$



c.

x-major
 $a = 8$

$\frac{x^2}{16^2} + \frac{y^2}{b^2} = 1$
 $\frac{8^2}{16^2} + \frac{6^2}{b^2} = 1$
 $\frac{64}{256} + \frac{36}{b^2} = 1$
 $\frac{36}{b^2} = 1 - \frac{64}{256}$
 $b^2 = 48$

Example 3: Find the standard equation for the ellipse that satisfies the given conditions.

- a. Foci: $(\pm 4, 0)$, vertices: $(\pm 5, 0)$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$a = 5$

$c = 4$

$4^2 = 5^2 - b^2$

$b^2 = 9$

$\frac{x^2}{25} + \frac{y^2}{9} = 1$

$\therefore \frac{x^2}{256} + \frac{y^2}{48} = 1$

b. Foci: $(0, \pm 2)$, $\xrightarrow{c}$ length of minor axis: 6

$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1$$

$$\boxed{\frac{y^2}{13} + \frac{x^2}{9} = 1}$$

$$2b = 6$$

$$b = 3$$

$$c^2 = a^2 - b^2$$

$$a^2 = c^2 + b^2$$

$$a = \sqrt{13}$$

c. Length of major axis: 4, length of minor axis: 2, foci on y-axis
 $\xrightarrow{y\text{-major}}$

$$a = 2$$

$$b = 1$$

$$\boxed{\frac{x^2}{1} + \frac{y^2}{4} = 1}$$

d. Endpoints of minor axis: $(0, \pm 3)$, $\xrightarrow{b}$ distance between foci: 8

$x\text{-major}$

$$\boxed{\frac{x^2}{25} + \frac{y^2}{9} = 1}$$

$$c - (-c) = 8$$

$$2c = 8$$

$$c = 4$$

$$a^2 = c^2 + b^2$$

$$= 16 + 9$$

$$a^2 = 25$$

e. Length of minor axis: 10, foci on y-axis, ellipse passes through the point $(\sqrt{5}, \sqrt{40})$.
 $\xrightarrow{y\text{-major}}$

$$b = 5$$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$\frac{\sqrt{5}^2}{25} + \frac{\sqrt{40}^2}{a^2} = 1$$

$$\frac{5}{25} + \frac{40}{a^2} = 1$$

$$\frac{1}{5} + \frac{40}{a^2} = 1$$

$$\frac{40}{a^2} = \frac{4}{5}$$

$$a^2 = 50$$

$$\boxed{\frac{x^2}{25} + \frac{y^2}{50} = 1}$$

f. Eccentricity: $\frac{\sqrt{5}}{3}$, $\xrightarrow{x\text{-major}}$ foci on x-axis, length of major axis: 12

$$e = \frac{c}{a}$$

$$\frac{c}{6} = \frac{\sqrt{5}}{3}$$

$$c = 2\sqrt{5}$$

$$b^2 = a^2 - c^2$$

$$= 36 - 20$$

$$= 16$$

$$\boxed{\frac{x^2}{36} + \frac{y^2}{16} = 1}$$

$$2a = 12$$

$$a = 6$$

Example 4: Find the intersection points of the pair of ellipses. Sketch the graphs of each pair of equations on the same coordinate axes, and label the points of intersection.

a.
$$\begin{cases} 4x^2 + y^2 = 4 \\ 4x^2 + 9y^2 = 36 \end{cases}$$

$$\begin{cases} \frac{x^2}{1} + \frac{y^2}{4} = 1 \\ \frac{x^2}{9} + \frac{y^2}{4} = 1 \end{cases}$$

$$(4x^2 + 9y^2) - (4x^2 + y^2) = 36 - 4$$

$$8y^2 = 32$$

$$y^2 = 4$$

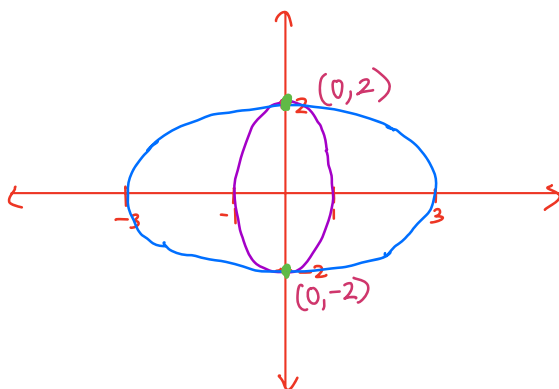
$$y = \pm 2$$

$$4x^2 + y^2 = 4$$

$$4x^2 + 4 = 4$$

$$x = 0$$

intersection pts: $(0, 2), (0, -2)$



b.
$$\begin{cases} \frac{x^2}{16} + \frac{y^2}{9} = 1 \\ \frac{x^2}{9} + \frac{y^2}{16} = 1 \end{cases}$$

$$-16(4x^2 + 16y^2 = 144)$$

$$9(16x^2 + 9y^2 = 144)$$

$$-144x^2 - 256y^2 = -2304$$

$$+ 144x^2 + 81y^2 = 1296$$

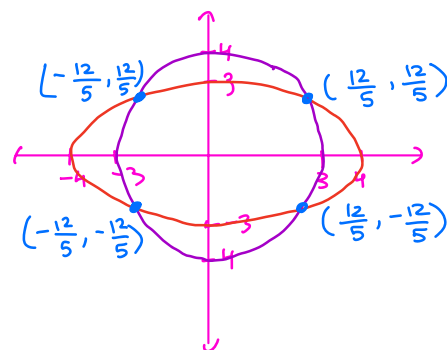
$$-175y^2 = -1008$$

$$y^2 = \frac{144}{25}$$

plug into
either
equation
above to
find x

$$y = \pm \frac{12}{5}$$

$$x = \pm \frac{12}{5}$$



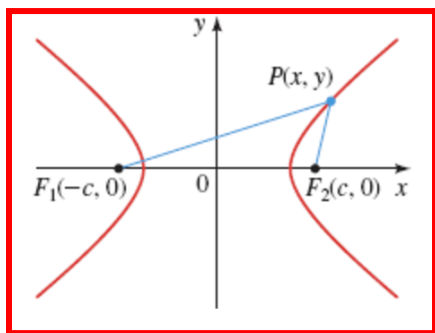
Chp 11.3: Hyperbolas

Although ellipses and hyperbolas have completely different shapes, their definitions and equations are similar. Instead of using the **sum** of distances from two fixed foci, as in the case of an ellipse, we use the **difference** to define a **hyperbola**.

Geometric Definition of a Hyperbola

A **hyperbola** is the set of all points in the plane, the **difference** of whose **distances** from **two** fixed points F_1 and F_2 is a **constant**. These two fixed points are the **foci** of the hyperbola.

P is on the hyperbola if $|d(P, F_1) - d(P, F_2)| = 2a$.



$$x^2 - 2y^2 + \dots$$

The **standard equation** of a hyperbola with foci on the x -axis and center at the origin is

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If we replace x by $-x$ or y by $-y$ in this equation, it remains unchanged, so the

hyperbola is symmetric about **both the x - and y -axes and about the origin**. The **x -intercepts** are **$\pm a$** , and the points **$(a, 0)$ and $(-a, 0)$** are the **vertices** of the hyperbola. There is no

y -intercept, because setting $x = 0$ in the equation of the hyperbola leads to $-y^2 = b^2$ which has no real solution. The hyperbola consists of **two parts**, called its **branches**. The segment joining the two vertices on the separate branches is the **transverse axis** of the hyperbola, and the origin is called its **center**.

If we place the foci of the hyperbola on the y -axis rather than on the x -axis, this has the effect of **reversing the roles of x and y** in the derivation of the equation of the hyperbola. This leads to a hyperbola with a vertical transverse axis.

Hyperbola with Center at the Origin

The graph of each equation is a hyperbola with center at the origin and having the given properties.

EQUATION

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (a > 0, b > 0) \quad \text{→ x-major}$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 \quad (a > 0, b > 0) \quad \text{→ y-major}$$

VERTICES

$$(\pm a, 0)$$

$$(0, \pm a)$$

TRANSVERSE AXIS

Horizontal, length $2a$

Vertical, length $2a$

ASYMPTOTES

$$y = \pm \frac{b}{a}x$$

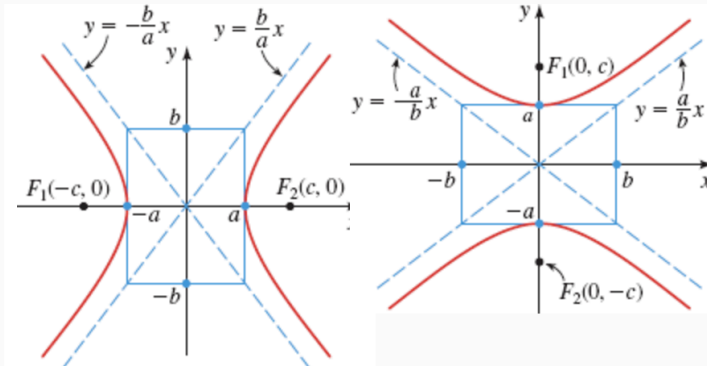
$$y = \pm \frac{a}{b}x$$

FOCI

$$(\pm c, 0), c^2 = a^2 + b^2$$

$$(0, \pm c), c^2 = a^2 + b^2$$

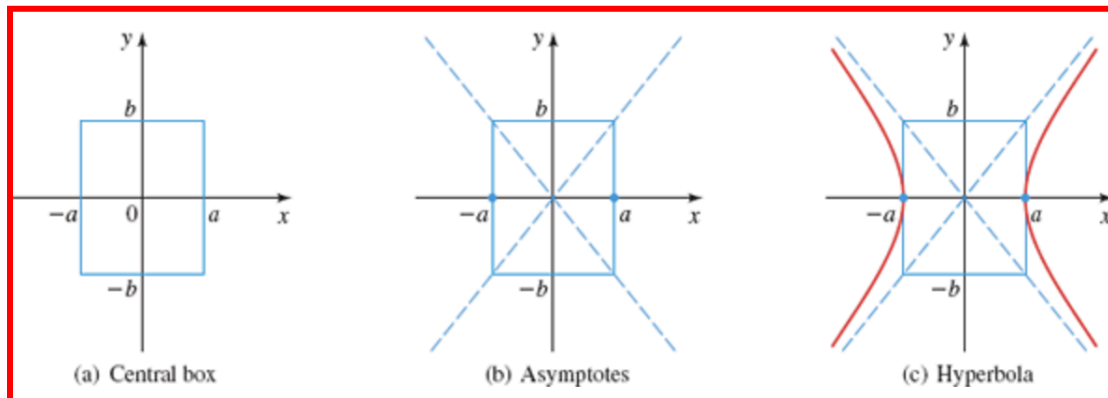
GRAPH



How to Sketch a Hyperbola

1. Sketch the **Central Box**. This is the rectangle centered at the **origin**, with sides parallel to the axes, that crosses one axis at $\pm a$ and the other at $\pm b$.
2. Sketch the **Asymptotes**. These are the lines obtained by **extending the diagonals** of the central box.
3. Plot the **Vertices**. These are the **two x-intercepts** or the **two y-intercepts**.
4. Sketch the **Hyperbola**. Start at a **vertex**, and sketch a branch of the **hyperbola**, approaching the **asymptotes**. Sketch the other branch in the same way.

Steps in graphing the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.



Example 1: An equation of a hyperbola is given.

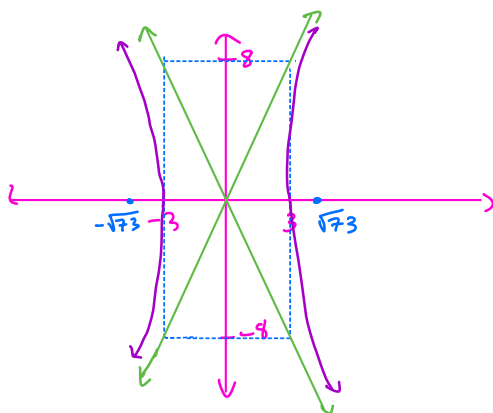
- Find the vertices, foci, and asymptotes of the hyperbola.
- Determine the length of the transverse axis. $\rightarrow$ major axis
- Sketch a graph of the hyperbola.

i. $\frac{x^2}{9} - \frac{y^2}{64} = 1$ $\rightarrow$ x-major
 $a=3$
 $b=8$
 $c^2 = a^2 + b^2$
 $c = \sqrt{64+9}$
 $= \sqrt{73}$

$(\pm\sqrt{73}, 0)$

$2a = 2 \cdot 3 = 6$

$y = \pm \frac{b}{a} x \rightarrow y = \pm \frac{8}{3} x$



ii. $\frac{y^2}{36} - \frac{x^2}{4} = 1$

y-major

$a=6$

$b=2$

$c^2 = a^2 + b^2$

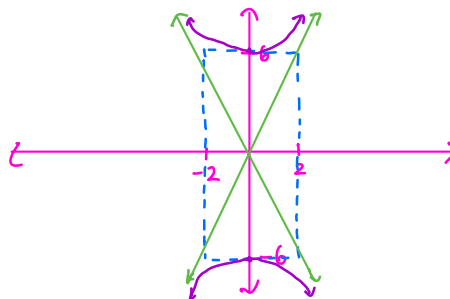
$c^2 = 36+4$

$c = \pm\sqrt{40}$

$(0, \pm\sqrt{40})$

$2a = 2 \cdot 6 = 12$

$y = \pm \frac{a}{b} x = \pm \frac{6}{2} x = \pm 3x \rightarrow y = \pm 3x$



iii. $x^2 - 3y^2 + 12 = 0$

$x^2 - 3y^2 = -12$
 $\frac{x^2}{-12} - \frac{3y^2}{-12} = 1$

$-\frac{x^2}{12} + \frac{y^2}{4} = 1$

$\frac{y^2}{4} - \frac{x^2}{12} = 1$

y-major

$a=2$

$b=\sqrt{12}$

$c^2 = a^2 + b^2$

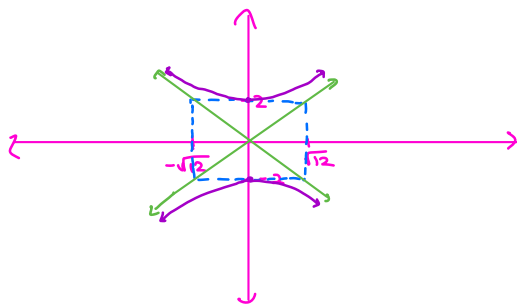
$= 4+12$

$= 16$

$c = \pm\sqrt{16}$

$2a = 2 \cdot 2 = 4$

$y = \pm \frac{a}{b} x = \pm \frac{2}{\sqrt{12}} x = \pm \frac{\sqrt{3}}{3} x \rightarrow y = \pm \frac{\sqrt{3}}{3} x$



iv. $9x^2 - 16y^2 = 1$ $\rightarrow$ x-major

$\frac{x^2}{\frac{1}{9}} - \frac{y^2}{\frac{1}{16}} = 1$

$a = \frac{1}{3}$

$b = \frac{1}{4}$

$c = \sqrt{\frac{1}{16} + \frac{1}{9}}$

$= \pm\sqrt{\frac{9+16}{144}}$

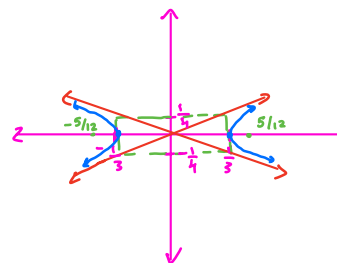
$= \pm\sqrt{\frac{25}{144}}$

$= \pm \frac{5}{12}$

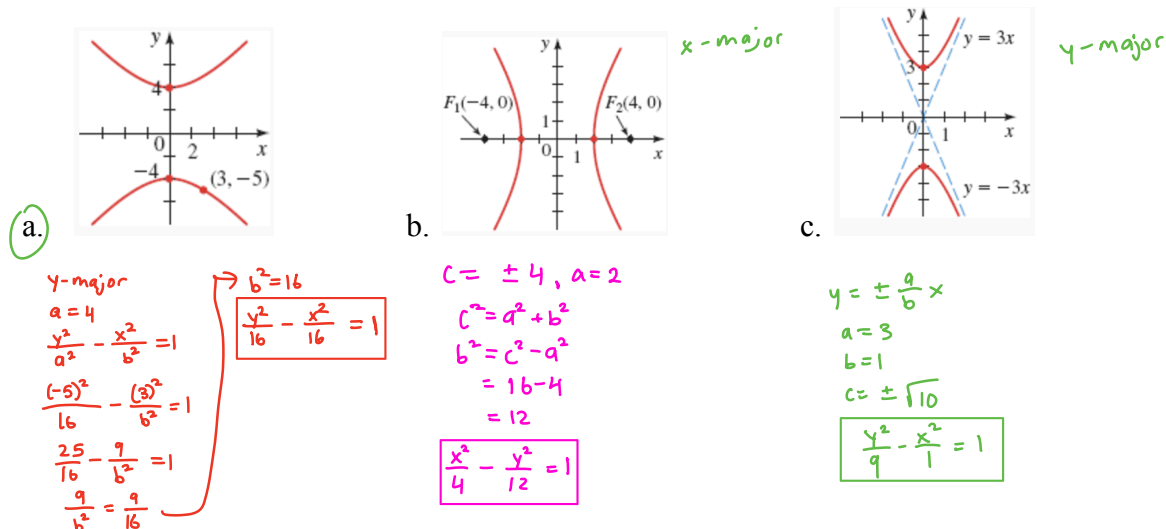
$y = \pm \frac{b}{a} x$
 $= \pm \frac{\frac{1}{4}}{\frac{1}{3}} x$

$y = \pm \frac{3}{4} x$

$2a = 2 \left(\frac{1}{3} \right) = \frac{2}{3}$



Example 2: Find the standard equation for the hyperbola whose graph is shown.



Example 3: Find the standard equation for the hyperbola that satisfies the given conditions.

- a. Foci: (± 6, 0) vertices: (± 2, 0)
- Handwritten work for (a):**
- x-major
 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
 $c = 6$
 $a = 2$
 $c^2 = a^2 + b^2$
 $6^2 = 2^2 + b^2$
 $b^2 = 32$
 $\frac{x^2}{4} - \frac{y^2}{32} = 1$
- b. Vertices: (± 1, 0), asymptotes: $y = \pm 5x$
- Handwritten work for (b):**
- x-major
 $a = 1$
 $y = \pm \frac{b}{a}x$
 $y = 5x$
 $\frac{b}{a} = 5$
 $\frac{b}{1} = 5$
 $b = 5$
 $\frac{x^2}{1} - \frac{y^2}{25} = 1$
- c. Asymptotes: $y = \pm x$, hyperbola passes through (5, 3)
- Handwritten work for (c):**
- $a = b$
 $c^2 = a^2 + b^2$
 $= a^2 + a^2$
 $c^2 = 2a^2$
 $\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$
 $\frac{25}{a^2} - \frac{9}{a^2} = 1$
 $\frac{16}{a^2} = 1$
 $a^2 = 16$
 $a = \pm 4 \rightarrow a = 4$
 $\frac{x^2}{16} - \frac{y^2}{16} = 1$

- d. Foci: (0, ± 1), length of transverse axis: 1

Handwritten work for (d):

y-major
 $2a = 1$
 $a = \frac{1}{2}$
 $\frac{y^2}{\frac{1}{4}} - \frac{x^2}{b^2} = 1$
 $c = 1$
 $c^2 = a^2 + b^2$
 $c^2 - a^2 = b^2$
 $b^2 = 1 - \frac{1}{4}$
 $= \frac{3}{4}$
 $\frac{y^2}{\frac{1}{4}} - \frac{x^2}{\frac{3}{4}} = 1$